PLANT SIGNALING & BEHAVIOR
2026, VOL. 21, NO. 1, 2687952
https://doi.org/10.1080/15592324.2026.2687952

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REVIEW ARTICLE

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Endophyte function in climate-stressed crops: integrating molecular regulation, metabolic trade-offs, and ecological constraints

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ABSTRACT

Climate change increasingly exposes crops to simultaneous abiotic and biotic stresses, disrupting physiological stability and reducing agricultural productivity. Although endophytes are widely recognized for improving plant stress tolerance, their effectiveness remains inconsistent across environmental conditions. This review develops an integrative framework to explain how endophyte-mediated responses are regulated at the molecular, metabolic, and ecological levels under climate stress. We examine multi-level interactions to inform predictive deployment in varied agricultural environments. We critically synthesize recent studies on ROS–hormone signaling, carbon allocation trade-offs, metabolic reprogramming, microbiome interactions, ecological filtering, and host genotype-dependent responses. We aim to identify mechanisms that govern the stability or destabilization of plant–endophyte associations under stress conditions. The analysis indicates that endophyte-mediated beneficial traits remain effective only when redox regulation, metabolic balance, and ecological compatibility are maintained within functional physiological limits. Under severe or prolonged stress, disruption of ROS homeostasis, carbon limitation, and ecological instability progressively reduce symbiotic effectiveness, physiological stability, and plant growth performance. Potential predictive variables in this framework include ROS accumulation thresholds, antioxidant capacity, photosynthetic stability, carbon allocation balance, and ecological persistence of endophytes under fluctuating environmental conditions. These variables may enable the prediction of endophyte functional stability, stress adaptation, and growth promotion under heterogeneous climate stress conditions. Collectively, this framework advances current understanding from descriptive interpretation toward a context-dependent perspective that may support future prediction and validation of endophyte performance in climate-resilient agriculture.

ARTICLE HISTORY
Received 13 May 2026
Revised 2 June 2026
Accepted 8 June 2026

KEYWORDS
Climate stress; ecological filtering; endophyte–plant interactions; host genotype; microbiome interactions; predictive deployment

1. Introduction

Climate change has become a major threat to agricultural productivity, driven by the increasing frequency, intensity, and co-occurrence of abiotic stressors such as drought, heat, salinity, and irregular precipitation, as well as biotic stressors, particularly plant diseases.[1,2] Recent climate datasets further indicate that climate change is accelerating, with 2025 ranked among the hottest years on record, according to the State of the Global Climate Update from the World Meteorological Organization (WMO), contributing to increasingly severe weather extremes that destabilize agricultural systems.[3] For instance, in rice, grain yield can decline by approximately 7% for every 1 °C increase in nighttime temperature above the optimum threshold of

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22 °C,[4] while heat stress during grain filling accelerates grain development, shortens the grain-filling duration, and compromises grain quality and final yield.[5,6] Drought can cause yield losses of up to 50% in major cereals, including wheat, maize, and rice, particularly when water deficits occur during flowering and grain filling, which are the most drought-sensitive stages of crop development.[7,8] Heavy rainfall, often accompanied by strong winds, can induce crop lodging (stem bending or breakage), resulting in substantial yield and quality losses. In severe cases, lodging has been reported to reduce wheat and other cereal yields by approximately 60%, particularly when it occurs during grain filling and before harvest.[9,10]

Plants are increasingly exposed to multiple, interacting stressors throughout their developmental cycle, with combined drought and heat among the most prevalent climate-related challenges.[11,12] Meta-analyzes indicate that heat and drought independently reduce crop yields by approximately 27% and 22%, respectively, whereas their combined occurrence can result in yield losses up to 49%, reflecting amplified physiological disruption and metabolic imbalance.[11] At the cellular level, combined stresses impair photosynthetic performance, disrupt metabolic homeostasis, and induce extensive molecular reprogramming, ultimately reducing plant growth, yield stability, product quality, and nutrient acquisition.[12,13] The increasing complexity and unpredictability of climate-induced stress highlight the urgent need for integrative climate-smart agricultural practices that maintain plant performance under dynamically fluctuating environmental conditions.

Endophytes are non-pathogenic bacteria and fungi that colonize internal plant tissues, including roots, stems, leaves, seeds, and vascular tissues, without causing disease symptoms.[14,15] Although many beneficial microorganisms have been reported, including rhizosphere-associated bacteria, arbuscular mycorrhizal fungi, *Trichoderma* spp., and other plant-associated microbes that contribute to plant stress tolerance, endophytes occupy a distinct ecological niche within host tissues. This internal lifestyle enables continuous exposure to host metabolites, signaling molecules, and stress-responsive pathways, allowing direct interaction with plant physiological, metabolic, and molecular regulatory processes.[16,17] The focus of this review is therefore not based on the assumption that endophytes are inherently superior to other beneficial microorganisms, but rather on the premise that their intimate association with internal plant tissues provides a unique framework for examining how molecular regulation, metabolic trade-offs, and ecological filtering collectively shape stress adaptation under climate change. Nevertheless, as with other beneficial microorganisms, the effectiveness of endophytes remains strongly dependent on host genotype, environmental conditions, and microbiome composition.[18,19]

Current understanding of endophyte function primarily focuses on beneficial outcomes, without fully considering the ecological, physiological, and climate-dependent factors that determine their consistency across environments. Under severe stress, endophyte colonization may not translate into improved growth, as plants prioritize physiological stability and survival over performance.[20,21] In contrast, under moderate stress, plant–endophyte responses can enhance stress tolerance and maintain functional performance.[22,23] However, this shift from moderate to severe stress results in a carbon-allocation trade-off, increasing investment in stress-adaptive metabolic pathways and thereby reducing the availability of assimilates for growth and yield formation.[19,24] These different outcomes highlight that endophyte effects are inherently conditional and shaped by system-level limitations.

Despite growing recognition of the multi-scale determinants of plant–endophyte interactions, a predictive and integrative framework remains limited. Existing studies commonly examine molecular regulation, metabolic responses, or ecological interactions separately, limiting their capacity to explain variability across complex environmental conditions. Furthermore, strong context-dependent factors driven by fluctuating temperature, soil properties, stress intensity, and biotic interactions are rarely resolved in single-factor studies. As a result, current models remain insufficient to predict the net effects of multiple, simultaneous, and dynamically interacting stressors under climate change.[25,26] A systems-level framework integrating molecular, metabolic, and ecological interactions is therefore necessary to determine when endophytes enhance plant performance and when their effects become limited. Potential predictive indicators within this framework may include ROS accumulation thresholds, antioxidant capacity, photosynthetic stability, carbon allocation balance, and ecological persistence of endophytes under fluctuating environmental conditions. These indicators may help define the functional conditions under which endophyte-mediated benefits remain stable.

As illustrated in Figure 1, this framework proposes a conceptual shift from outcome-based evaluation toward a context-dependent functional perspective regulated by physiological stability, resource availability, and ecological compatibility. Within this framework, the bounded functional range refers to the physiological and ecological conditions under which endophyte-mediated benefits remain stable and measurable. This range may be characterized by indicators such as ROS accumulation and antioxidant capacity, maintenance of photosynthetic performance and carbon allocation, persistence of endophyte colonization, microbiome compatibility, and positive plant growth or yield responses. Beyond these boundaries, increasing stress intensity may destabilize plant–endophyte interactions and reduce the consistency of beneficial outcomes.

The methodology employed in this review involves systematic identification and analysis of peer-reviewed literature published between 2019 and 2026, with particular emphasis on studies from the last five years addressing endophyte function in climate-stressed crops. Relevant articles were identified through Web of Science (https://webofknowledge.com) and Scopus (https://www.scopus.com) databases, using combinations of search terms such as "endophyte", "climate stress", "abiotic stress", "biotic stress", "ROS signaling", "phytohormones", "carbon allocation", "ecological filtering", "host genotype", and "microbiome interactions". Boolean operators were applied to refine and combine search terms to capture variations in terminology and conceptual overlap across molecular, metabolic, and ecological dimensions of plant–microbe interactions. Priority was given to experimental studies providing mechanistic insights into endophyte-mediated stress adaptation, particularly those addressing molecular regulation, metabolic trade-offs, and ecological constraints under climate-related stress conditions in crops. Additional review articles were included where necessary to contextualize emerging concepts and integrate cross-scale evidence. The selected literature was critically evaluated to synthesize current understanding, identify knowledge gaps, and support the development of an integrative framework for predicting endophyte function under climate stress.

In the present work, we synthesize current knowledge on plant–endophyte interactions across molecular, metabolic, and ecological scales and develop an integrative framework to explain their context-dependent outcomes. Specifically, it (i) elucidates regulatory constraints governing reactive oxygen species (ROS) signaling and phytohormonal interactions, (ii) examines metabolic limitations associated with

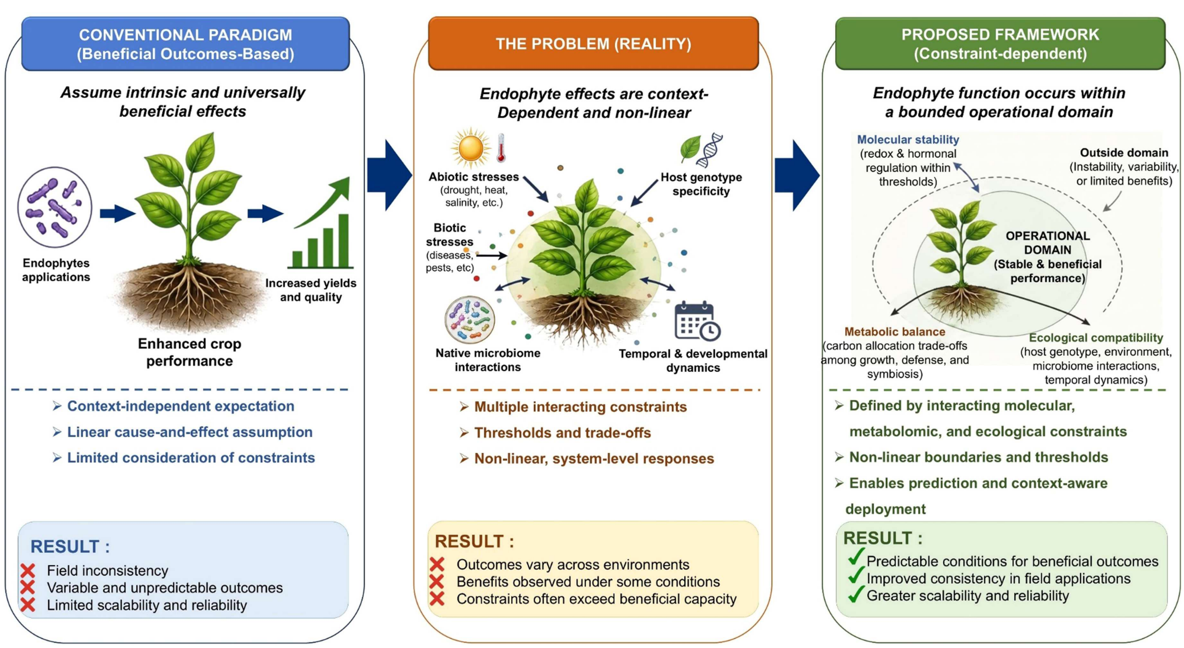

Figure 1. Framework shift from outcome-based evaluation toward a context-dependent framework of endophyte function under climate stress. The proposed framework defines a bounded operational domain in which endophyte-mediated benefits remain stable through molecular stability, metabolic balance, and ecological compatibility. Outside this domain, increasing stress and ecological constraints reduce the consistency of beneficial outcomes.

resource allocation under stress, and (iii) evaluates ecological filters that determine endophyte establishment and functional performance across environments. By aligning endophyte application with physiological limits and environmental context, this review advances a shift from descriptive interpretation toward a mechanistic and predictive paradigm, supporting more reliable deployment in climate-resilient crop production systems.

2. Molecular interactions: threshold-dependent regulation of ROS–hormone networks

2.1. Redox dynamics as a constrained signaling system

Endophyte-mediated modulation of plant stress responses is primarily determined by the biophysical and biochemical properties of cellular redox systems, which set the functional capacity of antioxidant networks and electron transport processes to maintain homeostasis under stress.[20,27] At the cellular level, ROS are continuously generated through electron leakage from chloroplastic and mitochondrial electron transport chains, as well as via plasma membrane–localized NADPH oxidases (respiratory burst oxidase homologs, RBOHs).[28,29] Under moderate stress, ROS act as essential signaling molecules that induce transcriptional reprogramming and metabolic adjustments required for stress acclimation.[30,31] This signaling function is maintained within a narrow redox range controlled by activity of enzymatic antioxidants such as superoxide dismutase (SOD), catalase (CAT), ascorbate peroxidase (APX), and peroxidase (POD), together with non-enzymatic components including ascorbate (AsA), glutathione (GSH), carotenoids, and proline.[27,32,33]

As shown in Figure 2, endophytes reinforce this regulatory capacity by enhancing antioxidant activity and priming ROS-dependent signaling, thereby maintaining ROS within functional signaling range under suboptimal but non-lethal stress conditions.[34,35] This regulation improves photosynthetic stability,

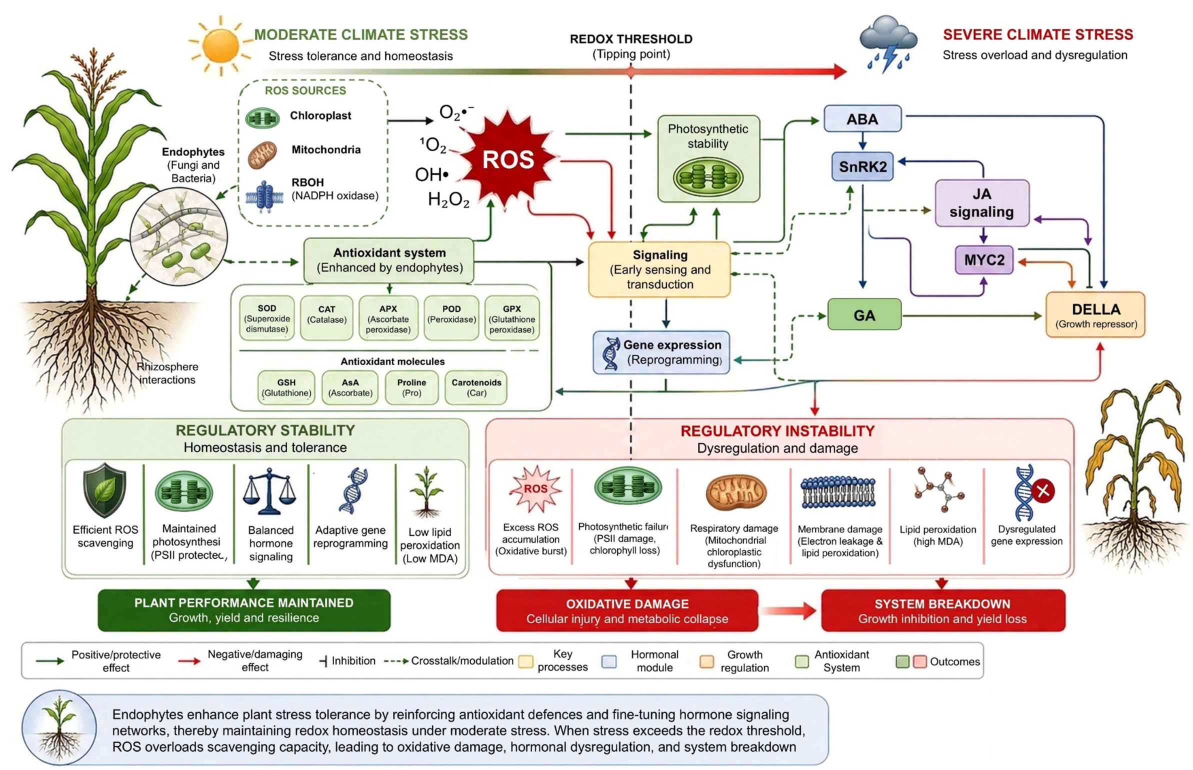

Figure 2. Endophyte-mediated redox regulation across increasing climate stress intensity. Under moderate stress, endophytes maintain ROS homeostasis through antioxidant defense mechanisms and hormone-mediated signaling, thereby supporting photosynthetic stability and adaptive responses. When stress exceeds the redox threshold, ROS accumulation overwhelms detoxification capacity, leading to oxidative damage, regulatory instability, and growth inhibition.

preserves photosystem II integrity, increases net photosynthesis, and promotes efficient signal transduction and adaptive gene expression.[18,36,37] In parallel, endophytes integrate redox signaling with key hormonal pathways, particularly the abscisic acid (ABA)–SnRK2, jasmonic acid (JA)–MYC2, and gibberellin (GA)–DELLA modules, thereby coordinating stress responses with developmental regulation.[38-40] Through modulation of ROS dynamics and hormonal balances, beneficial endophytes such as *Trichoderma* spp., *Serendipita indica* (Syn. *Piriformospora indica*), and *Bacillus* spp. enhance physiological stability by improving root architecture, nutrient uptake, and antioxidant defenses.[38,41,42]

When stress intensity increases, disruption of electron transport accelerates ROS accumulation beyond detoxification capacity, shifting the system from controlled signaling to oxidative imbalance.[43,44] As ROS exceed the redox threshold, antioxidant defense systems become insufficient to maintain homeostasis, stimulating oxidative damage and regulatory instability. Redox threshold exceedance can be experimentally assessed through coordinated increases in ROS and malondialdehyde (MDA) accumulation, declines in antioxidant enzyme activities, reductions in photosynthetic efficiency (Fv/Fm), and impaired plant growth performance.[27,45] This shift results in membrane destabilization, lipid peroxidation, increased MDA accumulation, and impairment of chloroplastic and mitochondrial function.[36,46] This redox imbalance also disrupts hormonal signaling networks, leading to altered gene expression, metabolic disruption, cellular damage, and growth inhibition.[36,46]

The limited effectiveness of endophyte-mediated protection under extreme conditions aligns with the Habitat-Adapted Symbiosis hypothesis, which proposes that symbiotic benefits are optimized within specific environments but decline under severe, non-native stress conditions.[47,48] Under severe environmental or physiological stress, the symbiotic relationship may become destabilized and, in some cases, shift from mutualistic to a less beneficial, or even parasitic, interaction.[20,49] This phenomenon is driven by a disruption in the association's cost-benefit balance, as activation of antioxidant systems and stress-responsive pathways requires substantial energy and carbon investment.[50,51] Consequently, increased allocation to protective functions reduces resources available for growth and yield formation. Endophytes, therefore, extend the functional range of redox regulation but remain limited by host physiological capacity.

2.2. Integrated ROS–hormone network dynamics and growth–defense trade-offs

Redox and phytohormonal pathways form an integrated regulatory network that controls plant responses to stress through coupled signaling and antioxidant processes.[52,53] ROS act as primary signals that induce hormone biosynthesis and activate stress-responsive transcriptional programs, while hormonal pathways, in turn, regulate ROS production through their effects on stomatal conductance, metabolic flux, and NADPH oxidase activity.[53,54] This bidirectional interaction establishes a regulatory system in which some processes maintain redox homeostasis, whereas others intensify stress signaling, thereby shaping the overall physiological response of the system.[55,56]

Endophytes, including fungi and bacteria, influence the ROS-hormonal regulatory network by coordinating ROS dynamics and phytohormonal signaling, thereby translating environmental stress signals into adaptive responses.[57-59] Key hormonal modules, including ABA–SnRK2, JA–MYC2, SA-mediated defense, and GA–DELLA pathways, mediate the translation of redox signals into coordinated physiological responses.[19,57] Rather than functioning independently, these pathways operate as an interconnected regulatory network that integrates redox and hormonal signals, defense activation, growth regulation, and physiological acclimation under adverse environmental conditions.

Under abiotic stress, endophytes regulate ABA signaling pathways that activate subclass III SnRK2 protein kinases involved in osmotic adjustment and stress-responsive gene expression.[60,61] However, activation of ABA-dependent pathways is energy-intensive and may reduce plant height, biomass accumulation, and root and shoot growth under prolonged stress conditions.[58,62] ABA signaling interacts closely with JA-mediated pathways, with the MYC2 transcription factor coordinating defense responses, secondary metabolism, and stress adaptation under both abiotic and biotic stresses.[59,63] Crosstalk between ABA–SnRK2 and JA–MYC2 modules coordinates redox regulation, defense activation, and physiological acclimation.[57,59] In addition, SA signaling strengthens pathogen defense through NPR1-mediated systemic acquired resistance (SAR).[64,65] SA signaling also interacts with ABA- and JA-regulated responses to maintain cellular homeostasis under environmental stress.[66,67] Recent studies are exploring endophyte-based strategies to optimize these interconnected pathways and enhance stress resistance without substantial yield penalties.

Endophytes additionally influence GA signaling by modulating GA homeostasis and DELLA-associated regulatory pathways, thereby balancing growth and stress adaptation under adverse environmental conditions.[68,69] Through the integration of ABA–SnRK2, JA–MYC2, SA, and GA–DELLA networks, endophytes regulate the balance between stress tolerance and growth, thereby strengthening resilience while reshaping developmental priorities under climate stress. A fundamental consequence of this integration is the emergence of growth–defense trade-offs. Activation of ABA- and JA-dependent pathways promotes stress tolerance, whereas stabilization of DELLA proteins suppresses GA-mediated growth.[70-72] Endophyte-mediated hormonal reprogramming operates by redistributing physiological priorities rather than by enhancing total system capacity.

This integrated ROS-hormone network is inherently non-linear and regulated by threshold-dependent dynamics.[19,73] Under moderate stress, regulatory interactions maintain system stability, enabling integrated responses that support survival while causing limited growth penalties. As stress intensifies, positive interactions between ROS accumulation and stress hormone signaling can drive excessive activation, overwhelm antioxidant defenses, and disrupt hormonal coordination.[44,57] The transition results in systemic instability characterized by oxidative damage, metabolic exhaustion, and loss of growth regulation. These contrasting responses indicate that plant stress adaptation is governed by stability boundaries rather than by linear changes in stress intensity.

From an integrative perspective, the ROS–hormone network can be conceptualized as a non-linear system defined by stability thresholds and feedback sensitivity.[19,73] Endophyte function is therefore conditional upon the system's proximity to these thresholds rather than scaling proportionally with stress intensity. Identifying how these boundaries vary across genotypes and environments is essential for predicting when endophyte-mediated regulation enhances plant performance and when its effects become limited or destabilizing. Potential measurable indicators associated with these stability thresholds include ROS accumulation, antioxidant enzyme activity (SOD, CAT, APX), photosynthetic efficiency, and hormone-responsive signaling dynamics (Table 1). These parameters provide a basis for future validation of threshold-dependent endophyte function under environmental stress.

3. Metabolic Constraints on Endophyte Function Under Environmental Stress

3.1. Carbon limitation and allocation trade-offs among growth, defense, and symbiosis

Environmental stress substantially disrupts plant carbon metabolism by impairing photosynthesis, respiration, and energy production.[88,89] Drought, salinity, heat, flooding, and pathogen-associated stress reduce chlorophyll stability, stomatal conductance, electron transport, and carbon fixation, thereby limiting assimilate production and metabolic capacity.[7,90] Damage to chloroplast structure and photosystem activity further constrains ATP and NADPH generation required for cellular maintenance and stress adaptation.[90-92]

Additionally, stress increases respiratory demand associated with protein turnover, membrane repair, ion transport, and defense metabolism.[93-95] Although plant mitochondria support ATP production during stress adaptation, mitochondrial function itself becomes increasingly vulnerable to oxidative and metabolic disruption under severe conditions.[96-98] The combined decline in carbon acquisition and rise in maintenance costs progressively restricts carbon availability for growth and reproduction.[99,100]

Carbon limitation also influences the stability of symbiotic interactions. Endophytes depend on host-derived carbon substrates to sustain colonization, metabolism, and persistence within plant tissues.[19,101] Experimental studies further demonstrate that environmental stress alters carbon fluxes and assimilate partitioning, thereby modifying the physiological conditions that regulate endophyte function.[102,103] Consequently, endophyte-mediated stress mitigation remains closely associated with host carbon status and respiratory demand, resulting in context-dependent symbiotic outcomes.[23,104]

Sufficient carbon availability enables relatively coordinated allocation among growth, defense, and symbiosis (Figure 3). However, progressive carbon limitation intensifies sink competition, shifting

Table 1. Representative studies on ROS–hormone regulatory pathways, associated endophyte species, key molecular markers, and plant responses under stress.

Regulatory module	Endophyte species	Key molecular markers	Major plant responses	References
ABA–JA signaling pathways	*Bacillus pumilus*, *Pseudomonas argentinensis*	ABA, JA, PYR/PYL, SnRK2, MYC2	ABA homeostasis, osmotic adjustment, activation of drought-responsive signaling	[59,74]
ABA-mediated phytohormone regulation	*Bacillus safensis*	IAA, ABA, tZ	Hormonal regulation, salt stress alleviation, improved plant growth and yield	[75]
Integrated hormones (ABA–SA–JA) signaling and ROS regulation	*Acinetobacter calcoaceticus*	PR1, SnRK2.5, LOX1, ACS3, proline	Coordinated growth–defense trade-offs, enhanced stress resilience, and improved tolerance to combined drought–pathogen stress	[61]
Microbiome–metabolite interaction network	*Delftia* sp., *Stenotrophomonas* sp.	Lipids, organic acids, phenylalanine metabolism, amino acid metabolism, tyrosine metabolism	Metabolic adjustment and adaptation to drought and rehydration stress	[76]
ROS–antioxidant and metabolite-regulating pathways	*Colletotrichum spaethianum*, *Gliocladiopsis aquaticus*	Chlorophyll, carotenoids, carbohydrates, proteins, polyphenols, flavonoids, proline, IAA	Enhanced photosynthetic performance, improved osmotic adjustment, increased metabolite accumulation, and enhanced drought tolerance	[77]
JA–SA and antifungal defense signaling pathways	*Bacillus subtilis*	JA, SA, defense-related enzymes, antifungal activity	Enhanced plant growth, induced systemic resistance, and improved resistance against Fusarium wilt	[64]
SA–NPR1 and antifungal defense signaling pathways	*Trichoderma longibrachiatum*	SA, PAL, NPR1, PR5, Defensin, total polyphenols, free radical scavenging activity	Enhanced systemic resistance, increased antioxidant defense, suppressed root rot infection, and improved plant growth	[65]
ROS–osmotic adjustment and photosynthetic regulation	*Serendipita indica*	CAT, POD, APX, MDA, proline, soluble sugars, Pn, Gs	Improved photosynthetic performance, osmotic adjustment, and drought tolerance	[78]
GA–DELLA signaling pathway	*Rhizoglomus intraradices*	GA_3, GA_4, GA_8, GA_9, GA_{15}, PtGA20ox, PtGA3ox, PtGA2ox, PtGaipb, PtGai	Regulated GA metabolism, enhanced antioxidant defense, promoted growth performance, and increased drought tolerance	[79]
ROS–antioxidant regulation	*Piriformospora indica*	MDA, SOD, POD, chlorophyll, Fv/Fm	Enhanced oxidative stress tolerance and maintenance of photosynthetic stability under drought and heat stress	[80]
Root microbiome dynamics and nitrogen-associated interactions	*Bradyrhizobium* sp. and diverse bacterial and fungal endophytes	16S rRNA, ITS, OTUs, ASVs, microbial diversity indices	Maintained root microbiome stability, enhanced microbial interactions, and supported plant growth across developmental stages	[81]
ABA–JA and ROS–antioxidant signaling pathways	*Lysinibacillus fusiformis*, *Lysinibacillus sphaericus*, *Brevibacterium pityocampae*	ABA, SA, JA, OsNHX1, OsAPX1, OsPIN1, OsCATA, OsSOS	Improved ion homeostasis, enhanced antioxidant defense, and increased salinity tolerance	[82]
OS–antioxidant regulatory network	*Piriformospora indica*	ROS, H_2O_2, SOD, CAT, APX, GR, ascorbate, glutathione, FAD genes	Enhanced oxidative stress tolerance and membrane stability under drought	[83]
ROS–antioxidant and ascorbate–glutathione cycle pathways	*Serendipita indica*	ROS, H_2O_2, MDA, SOD, APX, GR, AsA, GSH/GSSG	Enhanced antioxidant defense, accelerated ROS scavenging, maintained chlorophyll content, and increased drought tolerance	[38]
GA biosynthesis and GA–DELLA signaling pathway	*Fusarium* sp.	GAs, ent-kaurene diterpenoids	Enhanced seedling growth and regulated phytohormone-mediated development	[84]
ROS–antioxidant and salt-responsive signaling pathways	*Piriformospora indica*	Oxidoreductase activity, thioredoxin peroxidase, Rho-type GTPase, sterol biosynthesis genes	Enhanced oxidative stress protection, maintained cellular integrity, and improved salt stress adaptation	[85]
GA–JA and hormonal crosstalk signaling pathways	*Piriformospora indica*	ent-kaurene synthase, Aux/IAA genes, 12-oxophytodienoate reductase, receptor-like kinases	Enhanced hormonal signaling, improved cell wall remodeling, and increased salinity stress adaptation	[86]
GA–DELLA signaling pathway	*Bacillus amyloliquefaciens*	DELLA proteins, GA deactivation genes, SA, JA	Modulated GA signaling, regulated growth–defense	[68]

(*Continued*)

Table 1. (*Continued*)

Regulatory module	Endophyte species	Key molecular markers	Major plant responses	References
			balance, and improved stress adaptation under phosphate deficiency	
ROS–antioxidant and drought-responsive signaling pathways	*Bacillus amyloliquefaciens, Bacillus megaterium, Lysinibacillus fusiformis, Pseudomonas putida, Stenotrophomonas maltophilia*	Catalase, guaiacol peroxidase, photosystem II activity, stomatal function, relative water content	Enhanced antioxidant defense, maintained photosynthetic performance, improved water status, and increased drought tolerance	[87]

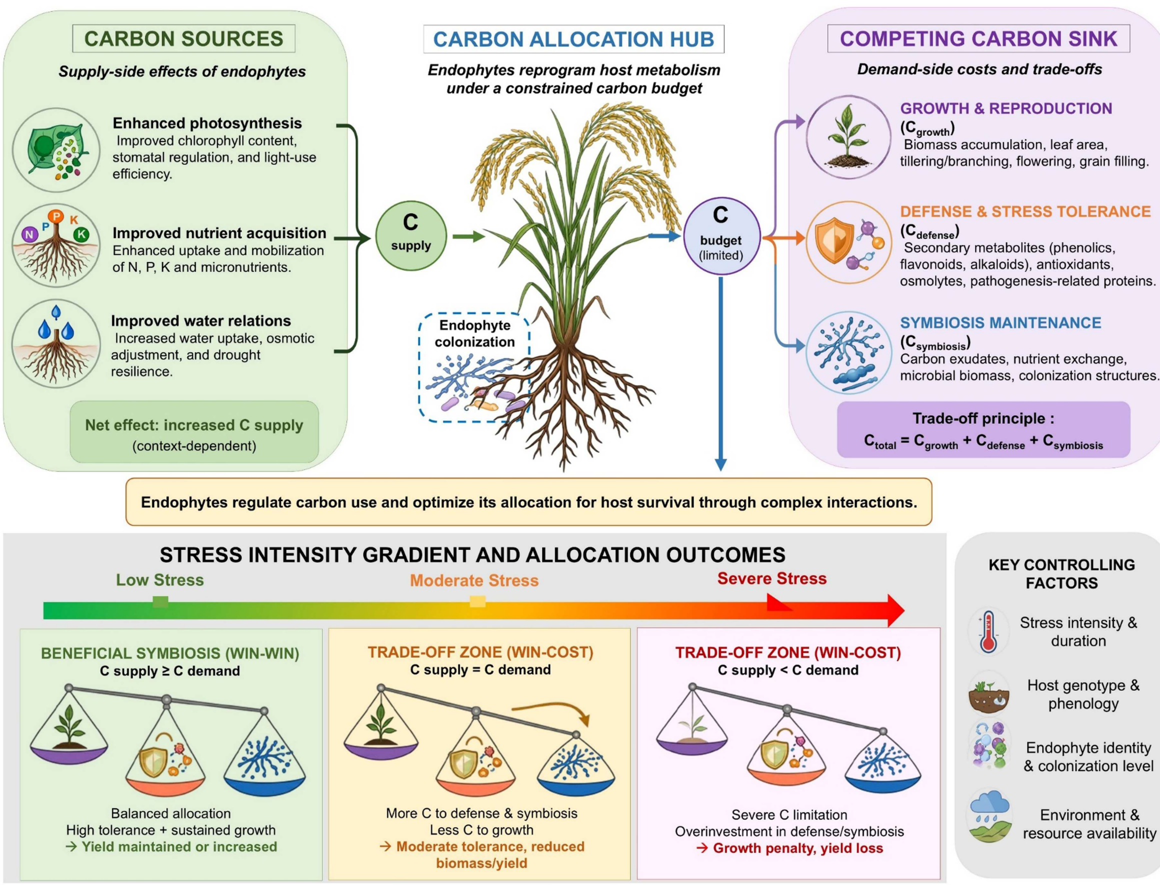

Figure 3. Carbon allocation trade-offs governing endophyte-mediated plant responses under climate stress. Endophytes influence carbon acquisition through effects on photosynthesis, nutrient uptake, and water relations while simultaneously altering carbon partitioning among growth, defense, and symbiotic maintenance. Increasing stress intensity progressively shifts carbon investment toward defense and symbiosis at the expense of growth and reproduction. The balance between carbon supply and demand ultimately determines whether plant–endophyte interactions sustain productivity or result in growth penalties and yield losses.

allocation toward survival-associated functions at the expense of vegetative growth and reproductive development. These responses represent adaptive metabolic prioritization within a constrained physiological system rather than pathological dysfunction.

3.2. Metabolic prioritization and source–sink competition

Stress adaptation requires extensive metabolic reprogramming that redirects cellular resources toward protective and survival-associated functions. Plants exposed to environmental stress increasingly allocate assimilates toward osmoprotectant synthesis, antioxidant metabolism, ion homeostasis, defense-related proteins, and cellular repair processes.[105-107] Accumulation of compatible solutes such as proline, soluble sugars, glycine betaine, and stress-associated secondary metabolites contributes to osmotic adjustment and cellular protection, but also increases metabolic expenditure during prolonged stress exposure.[108-110]

Defense-associated metabolism further elevates energetic demand by activating immune regulation, detoxification pathways, antioxidant systems, and the synthesis of protective metabolites.[111-113] Maintenance of endophytic associations may additionally require continuous allocation of carbon-derived metabolites, amino acids, and organic acids to support microbial persistence and metabolic activity within host tissues.[76,114,115] As multiple stress-responsive processes compete for assimilates, source–sink coordination becomes progressively restructured during stress adaptation.

Under these conditions, assimilate distribution increasingly shifts away from growth-associated and reproductive tissues toward immediate survival functions.[111,116,117] This metabolic prioritization helps sustain cellular integrity under stress but simultaneously reduces biomass accumulation and yield-related development.[116-118]

3.3. *Integration across scales: carbon economy as a determinant of endophyte function*

Carbon allocation provides the mechanistic interface through which molecular regulation is translated into physiological and agronomic outcomes.[119–121] Although signaling networks coordinate stress perception and response, the plant carbon economy ultimately determines how these responses are expressed across growth, defense, reproduction, and symbiotic maintenance.[117,119] Endophytes function within this framework by simultaneously modifying carbon acquisition, metabolic demand, and assimilate partitioning.

This perspective identifies the carbon economy as a systems-level determinant of endophyte function under environmental stress. Activation of stress-responsive pathways alone does not necessarily translate into sustained growth or yield maintenance because defense and stress adaptation require substantial metabolic investment. Consequently, the functional outcome of endophyte-mediated stress responses depends on whether carbon allocation remains sufficiently balanced to support both protection and developmental demand.

Framing plant–endophyte interactions through carbon allocation trade-offs therefore provides a predictive framework for understanding context-dependent symbiotic outcomes under climate stress. Within this systems perspective, endophyte performance is determined not only by the capacity to induce protective responses, but also by the ability to optimize assimilate distribution while minimizing growth and reproductive penalties associated with survival-oriented metabolism.

The effects of endophytes on the carbon economy can be evaluated using measurable physiological and metabolic indicators. These include photosynthetic rate, respiration rate, soluble sugar and starch concentrations, root exudation patterns, biomass allocation between shoots and roots, and yield-related traits such as grain number, grain weight, and harvest index.[122,123] These parameters provide a quantitative framework for assessing how endophytes influence carbon acquisition, allocation, and utilization under environmental stress, thereby enabling experimental validation of carbon-mediated trade-offs among growth, defense, and symbiotic maintenance (Table 2).

4. Ecological Filtering as a Determinant of Endophyte Function Across Environmental Gradients

Endophyte functional traits emerge through interconnected ecological filters that regulate colonization, persistence, and functional traits across environmental gradients[127,128] (Figure 4). Although ecological filters were originally defined as abiotic environmental constraints, the concept has been expanded in host-associated microbiome ecology to include host- and microbiome-associated filtering processes that govern microbial establishment, persistence, and function. Consequently, endophyte assembly is shaped by the combined influence of environmental conditions, host characteristics, and resident microbial communities. These filters, including host compatibility, environmental suitability, microbiome interactions, and developmental dynamics, act together to determine whether beneficial plant–microbe associations can be established and maintained.[19,21] Their integration shapes the conditions under which endophyte-mediated plant responses are consistently expressed across field environments (Table 3).

Table 2. Carbon-related physiological and metabolic indicators associated with endophyte-mediated stress adaptation in plants.

Indicator category	Measurable parameter	Biological significance	Predictive relevance	References
Carbon allocation and sugar transport	Sucrose, glucose, hexose levels, CoSWEET expression, leaf gas exchange, root colonization rate	Indicates endophyte-mediated carbon redistribution and enhanced photosynthetic activity	Associated with symbiotic efficiency and stress-resilient growth	[122]
Carbon metabolism and photosynthetic performance	Photosynthetic capacity, chlorophyll, carotenoids, carbohydrate metabolism genes, biomass	Maintenance of carbon assimilation and metabolic adjustment during drought	Serves as an indicator of drought adaptation and productivity stability	[124]
Photosynthetic carbon assimilation and osmotic adjustment	Photosynthetic pigments, net photosynthetic rate, stomatal conductance, transpiration rate, soluble sugars, soluble proteins, proline	Maintenance of carbon assimilation and osmotic balance under drought	Linked to drought resilience and metabolic persistence	[78]
Carbon metabolism and metabolic reprogramming	Organic acids, flavonoids, vitamin C, 2-oxoglutaric acid, citric acid, drought-responsive proteins	Endophyte-mediated metabolic adjustment and carbon reallocation under drought	Provides insight into metabolic resilience and adaptive stress performance	[125]
Photosynthetic performance and oxidative stability	Chlorophyll content, Fv/Fm, MDA, SOD, POD, survival rate	Maintenance of photosynthetic efficiency and cellular stability under stress	Combined drought and heat stress tolerance	[80]
Photosynthetic carbon assimilation and oxidative metabolism	Net photosynthetic rate, water use efficiency, chlorophyll index, ROS, ascorbate, glutathione	Coordination of carbon assimilation and redox homeostasis during drought	Serves as a marker of metabolic stability under drought	[126]
Oxidative metabolism and carbon maintenance	ROS, antioxidant enzymes, ascorbate, glutathione, fatty acid composition, FAD gene expression	Highlights antioxidant-mediated protection of metabolic and membrane functions	Indicates capacity for oxidative stress adaptation and carbon investment efficiency	[83]

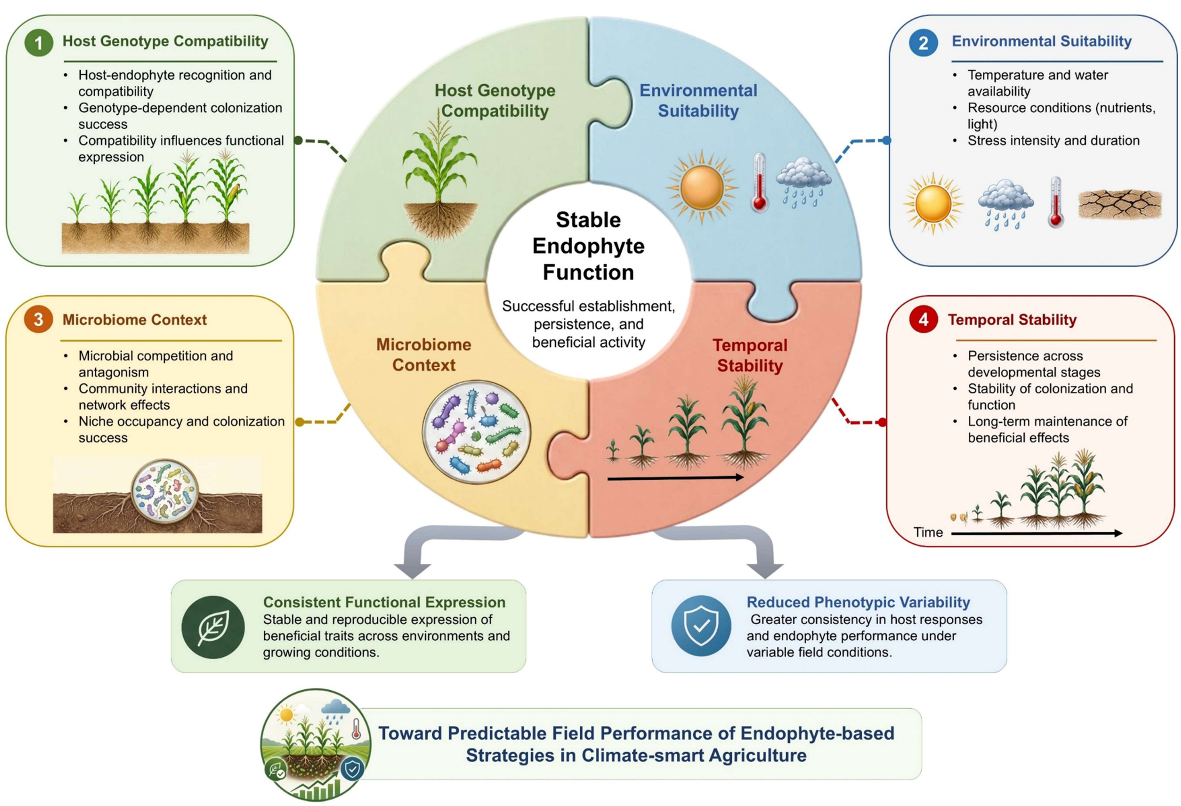

Figure 4. Ecological filters governing endophyte establishment and functional expression across environmental conditions. Host genotype compatibility, environmental suitability, microbiome context, and temporal stability collectively determine endophyte colonization success, persistence, and functional performance. The interaction among these filters influences the consistency of endophyte-mediated benefits and the reliability of plant responses under field conditions.

Table 3. Ecological filters shaping endophyte establishment, persistence, and functional variability under field conditions.

Ecological Filter	Key Determinant	Key Observations	Functional Interpretation	References
Host genotype × microbiome interaction	CsXPR1 gene regulating xylem microbial recruitment and colonization of *Pseudomonas fulva*	CsXPR1 expression level; ASV_4 abundance; colonization efficiency; plant growth traits (height, biomass, leaf area); metabolite (4-methyleneglutamine)	Genotype-dependent regulation of endophyte entry into vascular tissues determines microbial colonization success and translates into growth promotion and metabolic reprogramming	[129]
Host compatibility × microbial competition	Domain-dependent functional specialization of chitinases regulating colonization and inter-microbial interactions	Domain variation (CBM5); differential induction during colonization vs antagonism; immune suppression vs antifungal activity assays	Functional plasticity enables a trade-off between host immune suppression and microbial antagonism, determining colonization success and competitive outcome in the root environment	[130]
Environmental suitability × spatial heterogeneity	Edge-driven environmental filtering modulating fungal colonization within a single clonal genotype	ITS metabarcoding; epiphyte/endophyte diversity; community turnover along edge–interior gradient; taxonomic shifts (Dothideomycetes → Leotiomycetes)	Spatial environmental gradients override host genetic uniformity, increasing filtering intensity at forest edges and reshaping endophyte assembly independently of genotype	[131]
Environmental heterogeneity and microbiome composition	Habitat conditions, climate, and soil physicochemical properties	Endophytic bacterial and fungal communities differed significantly among habitats; community structure correlated with environmental variables and fruit quality traits	Endophyte establishment and functional expression are shaped by local environmental conditions and microbiome assembly processes	[128]
Temporal ecological stability	Seasonal variation and environmental conditions	Endophyte composition and assembly processes varied among seasons	Seasonal succession determines endophyte establishment and long-term stability	[132]
Host identity and environmental gradients	Host species and salinity conditions	Endophyte composition differed across hosts and salinity levels	Host- and environment-driven filtering determine which endophytes can colonize and persist	[21]
Environmental suitability × host genotype × soil-driven filtering	Soil properties have a stronger influence than host genotype on nodule microbiome assembly	16S rRNA diversity; soil type, pH, and nutrient status; nodule vs bulk soil diversity; taxon shifts	Soil factors dominate microbial assembly, producing specialized nodule communities and altering host recruitment under domestication	[133]
Genotype × environment × seed microbiome filtering	Host genotype and geographic location are structuring seed endophytic bacterial and fungal communities	16S/ITS diversity; ASVs; bacterial vs fungal diversity indices	Genotype–location interactions shape seed microbiome assembly, and microbial diversity is associated with variation in grain/malt quality traits	[134]
Microbiome competition and niche exclusion	Pre-existing microbial community structure	Endophyte inoculation altered microbial diversity and community composition, enriching specific beneficial taxa	Resident microbiome structure influences recruitment and establishment of introduced endophytes	[135]
Environmental + dispersal filtering in plant invasion context (alien host-endophyte system)	spersal mode (vertical vs horizontal) and environmental/biogeographic constraints across invasion stages	Seed-borne vs community-acquired endophytes; dispersal limitation; environmental filtering; host filtering	Ecological filtering across invasion stages governs endophyte acquisition and interaction outcomes, ranging from mutualism to pathogenicity under new environments.	[136]
Host immune signaling regulation	Host cell death regulation and TIR-NLR signaling	dAdo-induced cell death, TIR-NLR (ISI) activity, and fungal colonization level	Host immune pathways regulate endophyte colonization and symbiosis establishment	[137]
Host immune signaling regulation	Ferroptosis-mediated immune regulation during endophyte colonization	Ferroptosis markers, immune-related markers, endophyte colonization level, plant growth response, disease resistance	Controlled immune-associated cell death regulates symbiosis establishment and	[138]

(*Continued*)

Table 3. (*Continued*)

Ecological Filter	Key Determinant	Key Observations	Functional Interpretation	References
			endophyte-mediated benefits	
Host genotype-dependent microbiome selection	Plant genotype under combined water and phosphorus limitation	Distinct potato genotypes recruited different root-associated microbiomes and stress-responsive microbial functions	The host genotype acts as a primary ecological filter shaping microbiome assembly and stress-mitigation potential	[139]
Temporal ecological stability	Plant developmental stage and seasonal succession	Endophyte diversity and biomarker taxa shifted across growth stages	Temporal turnover influences endophyte persistence and functional consistency	[81]
Developmental stage dependency	Plant developmental stage	Endophyte effects on community assembly and network complexity varied across growth stages	Developmental transitions alter endophyte function and microbiome interactions over time	[140]
Microbiome competition and niche exclusion	Pre-existing microbial community structure and niche occupancy	Endophyte diversity, community composition, and assembly patterns varied across developmental stages and microbial backgrounds	Resident microbiomes can influence subsequent endophyte recruitment and community assembly	[81,140]
Temporal ecological stability	Plant developmental stage and field management practices	Endophyte composition changed across growth stages and cropping seasons	Developmental transitions shape endophyte persistence and field-level functional outcomes	[141]

4.1. Host compatibility and environmental regulation of colonization and function

Endophyte establishment depends on host–microbe compatibility, which requires coordinated microbial entry, molecular recognition, signal exchange, and integration within plant tissues to facilitate stable, mutually beneficial relationships.[142,143] Plants selectively permit colonization via immune perception systems and associated signaling pathways that distinguish compatible symbionts from non-beneficial or pathogenic organisms.[143] This selectivity influences both colonization ability and the extent to which microbial functions are integrated into host physiological processes.

Host genotype is a major determinant of plant–endophyte interactions. Variations in root exudate composition, cell wall properties, and immune responsiveness influence microbial recruitment and niche accessibility.[139,144] These genotype-dependent traits affect both colonization efficiency and functional integration, leading identical microbial strains to produce different responses across plant cultivars.[145]

Following colonization, environmental conditions regulate the functional activity of both the endophyte and the host.[134,146] Temperature, soil moisture, nutrient availability, and stress intensity directly influence microbial metabolism while simultaneously modifying host physiological responsiveness.[147,148] Functional expression, therefore, depends on the degree of environmental compatibility shared by both partners.

Under moderate stress conditions, compatible plant–endophyte interactions can maintain integrated physiological and metabolic responses that enhance nutrient acquisition, osmotic adjustment, antioxidant regulation, and overall plant performance.[149,150] As stress intensity increases, disruption of microbial metabolic activity, redox balance, and host physiological stability progressively limits functional expression even when microbial colonization persists within plant tissues.[76,151,152] In addition, spatial and temporal heterogeneity across field environments—including variation in soil moisture, nutrient distribution, temperature, and resident microbiome composition—can generate localized differences in endophyte activity and host responsiveness, thereby contributing to inconsistent outcomes across agricultural systems.[128,153,154]

4.2. Microbiome interactions and competitive constraints

Endophyte function is strongly influenced by interactions within the resident plant microbiome, where diverse microbial taxa simultaneously compete, coexist, and interact across limited ecological niches.[26,155,156] Introduced endophytes must therefore establish within pre-existing microbial communities that already occupy host tissues and compete for nutrients, colonization sites, and host-derived metabolites.[135,140] Under these conditions, colonization success depends not only on beneficial functional traits but also on ecological competitiveness, niche compatibility, and the capacity to persist within complex microbial networks. Consequently, endophyte strains that perform effectively under controlled conditions may be constrained from establishing stable populations in natural or agricultural environments.

Microbiome composition further modulates the direction, stability, and magnitude of endophyte-mediated responses. Cooperative interactions among microbial taxa can enhance nutrient acquisition, stress adaptation, metabolic complementarity, and host protection, whereas antagonistic interactions may suppress colonization, disrupt microbial signaling, or reduce functional expression.[35,157] Host plants actively regulate these interactions through immune selection, root exudation, and habitat filtering, thereby continuously reshaping microbiome structure across developmental stages and environmental conditions.[128,158,159] Endophyte performance, therefore, emerges from dynamic ecological interactions within the broader microbiome rather than from the intrinsic properties of individual strains alone.

4.3. Developmental dynamics and temporal variability

Endophyte–plant interactions vary throughout the plant life cycle as host physiological priorities shift from establishment to vegetative growth and reproduction.[81,140,160] These developmental transitions influence both microbial activity and the functional relevance of the symbiotic relationship.

During early plant development, endophytes contribute significantly to seed germination, root system establishment, and nutrient acquisition, thereby enhancing seedling vigor and early plant performance.[14,161,162] At this stage, endophyte–plant interactions are strongly associated with the modulation of root architecture, improved nutrient uptake efficiency, and enhanced physiological readiness for autotrophic growth, which collectively determine successful establishment under variable environmental conditions.[14,161]

During active vegetative growth, endophyte functions shift toward maintaining host physiological stability under abiotic stress conditions by regulating hormonal balance, enhancing antioxidant systems, and optimizing water and nutrient use efficiency.[139,163,164] These interactions are mediated by cooperative microbial networks that buffer stress-induced physiological disruption and sustain plant growth performance under suboptimal environments.[15,165]

The transition from vegetative to reproductive development represents a critical shift in plant–endophyte interactions. During this phase, the host increasingly reallocates carbon and assimilates toward flowering and fruiting organs.[148,166,167] This redistribution reduces the resources available for vegetative growth and reshapes whole-plant source–sink relationships. As a result, the functional contribution of endophytes is altered, leading to adjustments in microbial metabolic activity and interaction dynamics.[167,168] As reproductive development progresses, stronger host regulatory control increasingly constrains endophyte-associated functions, shifting their roles from broadly growth-supportive activities toward more specialized metabolic interactions associated with reproductive-stage physiology.[137,167,169]

Consequently, functional outcomes may change over time according to the host's developmental stage and physiological demands. Benefits observed during early growth phases may weaken, stabilize, or shift during later stages, contributing to differences between short-term experimental observations and season-long field performance.

4.4. Ecological filtering as a determinant of field inconsistency

Field performance of endophytes is ultimately shaped by multiple interacting ecological filters operating simultaneously across environmental conditions.[170,171] Host compatibility, microbiome competition, developmental timing, and environmental heterogeneity collectively determine whether introduced endophytes can establish stable associations and maintain functional activity within plant tissues.[21,172] Instability in any of these ecological processes can reduce colonization persistence and alter the magnitude of plant responses under field conditions.[172,173]

Environmental mismatch between controlled experimental systems and heterogeneous agricultural environments further contributes to translational variability. Endophytes demonstrating beneficial effects under laboratory or greenhouse conditions frequently encounter fluctuating environmental parameters, diverse microbial communities, and variable host responses that alter their functional performance after field deployment.[156,174] Consequently, ecological filtering is a major determinant of endophyte variation across agricultural systems.

These findings indicate that reliable application of endophytes in climate-stressed agriculture requires consideration of ecological compatibility in addition to microbial functionality. Stable field performance depends on the capacity of endophytes to persist within dynamic ecological environments characterized by host selectivity, microbiome interactions, and environmental variability. Understanding these ecological constraints is therefore essential for improving the predictability and translational success of endophyte-based strategies under field conditions.

5. Predictive endophyte deployment and future perspectives under climate change

The effectiveness of endophytes under climate stress depends on the coordinated stability of molecular regulation, metabolic balance, and ecological compatibility within the plant system. This perspective indicates that endophyte-mediated plant responses are not fixed microbial traits, but context-dependent outcomes shaped by stress intensity, host physiology, and environmental variability. Consequently, the successful deployment in climate-stressed agriculture requires a structured decision framework that moves from candidate selection to field validation under defined functional criteria.

At the screening stage, candidate endophytes should be selected based on functional traits related to stress mitigation, including antioxidant capacity, phytohormone modulation, carbon-use efficiency, and ecological adaptability, together with compatibility to specific host genotypes. At the testing stage, controlled environment assays should evaluate colonization stability, ROS regulation, metabolic allocation patterns, and interaction with native microbiomes under defined abiotic and biotic stress combinations. At the validation stage, multi-location field trials are required to assess persistence, functional consistency, and performance across heterogeneous soil, climatic, and microbiome conditions, with emphasis on stress specificity and genotype-dependent responses.

Within this framework, plant–endophyte interactions shift across functional states as environmental stress increases. Under moderate stress, coordinated redox regulation, carbon allocation, and ecological integration support adaptive responses. In this condition, endophytes can improve antioxidant regulation, nutrient acquisition, osmotic adjustment, and physiological stability, thereby contributing to relatively consistent plant performance. However, as stress intensity increases, these coordinated processes become increasingly unstable to fluctuations in temperature, water availability, nutrient status, and microbiome composition, resulting in stronger dependence on host genotype and environmental context.

This threshold-dependent behavior provides a mechanistic explanation for the translational gap frequently observed between controlled experiments and field performance. Laboratory and greenhouse systems typically maintain relatively stable plant–endophyte interactions. By stabilizing variables like humidity, light, and temperature, these settings allow endophytes to consistently synthesize beneficial metabolites, boost nutrient uptake, and enhance plant resilience without the disruptive fluctuations typical of field conditions.[156] In contrast, field environments expose plants to dynamically fluctuating stress combinations, heterogeneous soils, and complex microbial communities that continuously reshape symbiotic performance.[13,171] Variability in endophyte effectiveness, therefore, reflects the inherent sensitivity of plant–microbe systems to interacting environmental and physiological constraints rather than inconsistency in microbial potential alone.

Framing endophyte function within a predictive perspective has important implications for climate-resilient agriculture. Future endophyte deployment strategies should prioritize compatibility among host genotype, microbial functional traits, environmental conditions, and microbiome structure rather than assuming broad applicability across production systems. Integrating physiological analysis, microbiome profiling, transcriptomics, metabolomics, and environmental monitoring may improve the identification of functional indicators associated with stable endophyte performance under stress conditions.[76,175,176] Future validation of this framework may integrate physiological, metabolic, microbiome, and environmental indicators to identify the functional boundaries associated with stable endophyte performance across heterogeneous field conditions.

Advances in computational systems biology, ecological modeling, and machine learning further provide opportunities to predict endophyte functionality across heterogeneous environments by integrating multi-scale biological and environmental datasets.[177-179] In parallel, the development of stress-adapted microbial consortia may improve ecological persistence and functional complementarity under fluctuating field conditions compared with single-strain inoculants.[180,181] Such approaches may enhance the stability of plant responses under increasingly complex climate-associated stress environments.

Long-term validation across diverse agroecological systems remains essential to determine the environmental and physiological boundaries within which endophyte-mediated benefits can be consistently maintained. Future research should therefore increasingly focus on identifying the regulatory, metabolic, and ecological conditions that support stable plant–endophyte function under climate stress. This transition from descriptive evaluation to predictive deployment frameworks will be critical to improving the reliability of endophyte-based strategies in sustainable, climate-resilient agriculture.

6. Conclusions

Endophyte function in climate-stressed crops is determined by the coordinated interaction of molecular regulation, metabolic trade-offs, and ecological constraints that collectively define whether plant–microbe associations remain functional under fluctuating environmental conditions. Molecular regulation governs stress perception and signaling homeostasis, metabolic trade-offs determine the allocation of limited carbon and energy between growth and defense, while ecological constraints such as host selectivity, microbiome compatibility, and environmental heterogeneity ultimately regulate persistence and field-level effectiveness. Unlike controlled conditions, field environments impose simultaneous, multi-layered pressures that limit the stability of these interactions, thereby explaining the observed variability in endophyte performance. This review advances current understanding by integrating these three regulatory dimensions into a unified functional perspective, distinguishing it from prior studies that largely consider endophyte effects in isolation or under single-stress conditions. Future research should prioritize (i) defining measurable physiological and molecular thresholds that determine the transition from beneficial to constrained symbiosis, (ii) validating endophyte performance under multi-site field conditions with heterogeneous stress profiles, (iii) developing predictive models linking host genotype–microbe–environment interactions, and (iv) establishing standardized criteria for endophyte selection and deployment in climate-resilient agriculture.

Acknowledgments

Not applicable.

Author contributions

CRediT: **Wiwiek Harsonowati:** Conceptualization, Data curation, Investigation, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing; **Liauw Lia Sanjaya:** Data curation, Investigation, Writing – original draft, Writing – review & editing; **Amik Krismawati:** Investigation, Writing – original draft; **Janne Herly Willy Rembang:** Investigation, Writing – original draft; **Jefny Bernedi Markus Rawung:** Data curation, Investigation, Writing – original draft, Writing – review & editing; **Wahyu Widiyono:** Data curation, Investigation, Writing – original draft, Writing – review & editing; **Febri Doni:** Investigation, Writing – review & editing; **Rashid Iqbal:** Investigation, Writing – review & editing; **Sajid Ullah:** Investigation, Writing – review & editing.

Disclosure statement

The authors report there are no competing interests to declare.

Funding

No funding was received to assist with the preparation of this manuscript.

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Data availability statement

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

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